

## MP352 – Special Relativity

### Problem Set 4

Due by the end of lecture on Friday, 24 April 2015

1. Let  $\mathcal{S}$  and  $\mathcal{S}'$  be two inertial frames, with  $\mathcal{S}'$  moving relative to  $\mathcal{S}$  with a speed  $v$  in the positive  $x$ -direction.

- (a) Two 4-vectors  $A^\mu$  and  $B_\mu$  have the following components in  $\mathcal{S}$ :

$$\begin{aligned}(A^0, A^1, A^2, A^3) &= (2, 1, -1, 0), \\ (B_0, B_1, B_2, B_3) &= (-4, q, 1, 2).\end{aligned}$$

First, classify  $A^\mu$  as spacelike, timelike or lightlike. Then, determine the values of  $q$  for which  $B_\mu$  is spacelike, timelike or lightlike.

- (b) Find  $A'^\mu$  and  $B'_\mu$ , i.e. the components of the two 4-vectors in  $\mathcal{S}'$ .
  - (c) Using your results from (b), confirm that

$$||A'||^2 = ||A||^2, \quad ||B'||^2 = ||B||^2, \quad A'^\mu B'_\mu = A^\mu B_\mu.$$

2. We can describe any electromagnetic system by specifying the scalar potential  $\Phi(t, \vec{r})$  and the vector potential  $\vec{A}(t, \vec{r})$  and then finding the electric and magnetic fields using

$$\vec{E} = -\vec{\nabla}\Phi - \frac{\partial \vec{A}}{\partial t}, \quad \vec{B} = \vec{\nabla} \times \vec{A}.$$

It turns out that these potentials are actually the components of a 4-vector  $A^\mu$ , called the 4-potential, such that

$$A^0 = \frac{\Phi}{c}, \quad A^1 = A_x, \quad A^2 = A_y, \quad A^3 = A_z.$$

- (a) Using the 4-potential, we can define a new quantity, called the *field strength*  $F_{\mu\nu}$ , by

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Show that

$$\begin{aligned}F_{01} &= -E_x/c, & F_{02} &= -E_y/c, & F_{03} &= -E_z/c, \\ F_{12} &= B_z, & F_{23} &= B_x, & F_{31} &= B_y.\end{aligned}$$

- (b) The scalar and vector potentials are not unique, and as a result, we can always find  $\Phi$  and  $\vec{A}$  which satisfy the condition

$$\frac{1}{c^2} \frac{\partial \Phi}{\partial t} + \vec{\nabla} \cdot \vec{A} = 0.$$

Rewrite this condition using the 4-potential and the 4-derivative  $\partial_\mu$ , and explain why the expression you get means that if this condition is satisfied in any one inertial frame, it's automatically satisfied in *all* inertial frames.

3. You all (hopefully) know that moving charges create magnetic fields. Well, believe it or not, this is because of special relativity, and we're now going to show this. Suppose we start in an inertial frame  $\mathcal{S}'$  where a charge  $q$  is at rest at the origin. In this frame, the scalar and vector potentials are simply those of a stationary point charge, namely,

$$\Phi'(t', x', y', z') = \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{(x')^2 + (y')^2 + (z')^2}}, \quad \vec{A}'(t', x', y', z') = \vec{0}.$$

Now suppose  $\mathcal{S}'$  moves at a constant speed  $v$  in the  $x$ -direction relative to another frame  $\mathcal{S}$ . Thus, an observer at rest in this new frame sees the charge moving with constant velocity  $v\hat{e}_x$ .

- (a) Using the fact that  $A'^\mu$  is a 4-vector, find the scalar and vector potentials  $\Phi(t, x, y, z)$  and  $\vec{A}(t, x, y, z)$  in  $\mathcal{S}$ . (Note that we want these functions in terms of the *unprimed* coordinates, not the *primed* one!)
- (b) Using the relations between the potentials and the electric and magnetic fields in the previous problem, use your results from (a) to find the electric and magnetic fields as seen in  $\mathcal{S}$ . (Note that whereas there is *no* magnetic field in  $\mathcal{S}'$  where the charge is stationary, there *is* one in  $\mathcal{S}$  where the charge moves; Ampère was right!)
- (c) **EXTRA CREDIT** (not hard, but rather tedious): show that the electric and magnetic fields you found in (b) satisfy Maxwell's equations, remembering that  $\rho = 0$  and  $\vec{J} = \vec{0}$  for a point charge if you're not right on top of it. (And remember that  $\mu_0\epsilon_0 = 1/c^2$ .)

## MAXWELL'S EQUATIONS

$$\begin{aligned}\vec{\nabla} \cdot \vec{E} &= \rho/\epsilon_0, & \vec{\nabla} \cdot \vec{B} &= 0, \\ \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t}, & \vec{\nabla} \times \vec{B} &= \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}.\end{aligned}$$

## VECTOR CALCULUS FORMULAE IN CARTESIAN COORDINATES

- gradient of a function  $f(x, y, z)$ :

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{e}_x + \frac{\partial f}{\partial y} \hat{e}_y + \frac{\partial f}{\partial z} \hat{e}_z$$

- divergence of a vector  $\vec{A}(x, y, z) = A_x(x, y, z)\hat{e}_x + A_y(x, y, z)\hat{e}_y + A_z(x, y, z)\hat{e}_z$ :

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

- curl of a vector  $\vec{A}(x, y, z) = A_x(x, y, z)\hat{e}_x + A_y(x, y, z)\hat{e}_y + A_z(x, y, z)\hat{e}_z$ :

$$\vec{\nabla} \times \vec{A} = \left( \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{e}_x + \left( \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{e}_y + \left( \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{e}_z$$

- Laplacian of a function  $f(x, y, z)$ :

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$